Matrices from Rules

Exact construction, symbolic algebra, and limits

A matrix rule evaluates one expression over one-based row and column indices.

An exact Hilbert matrix

Define the $n \times n$ Hilbert matrix by

$$H(n) = \left(\frac{1}{i + j - 1} \right)_{i,j}.$$

At $n = 3$, the rule expands to

$$H\left(3\right) = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}$$

Its determinant stays exact:

$$\begin{aligned} H\left(3\right) &= \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix} \\ &\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix} \qquad \left(R_2 \leftarrow R_2 - \left(\frac{1}{2} \right) R_1 \right) \\ &\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{4}{45} \end{pmatrix} \qquad \left(R_3 \leftarrow R_3 - \left(\frac{1}{3} \right) R_1 \right) \\ &\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ 0 & 0 & \frac{1}{180} \end{pmatrix} \qquad \left(R_3 \leftarrow R_3 - \left(\frac{\frac{1}{12}}{\frac{1}{12}} \right) R_2 \right) \\ \det\left(H\left(3\right)\right) &= 1 \cdot \frac{\frac{1}{12}}{3} \cdot \frac{\frac{1}{180}}{5} \\ &= \frac{1}{2160} \end{aligned}$$

Gauss–Jordan elimination gives an integer inverse:

$$
\begin{aligned}
[H(3) \mid I] &= \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & 0 & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & 0 & 0 & 1 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & 1 & 0 & 0 \\ 0 & \frac{\frac{1}{4}}{3} & \frac{\frac{1}{3}}{4} & -\left(\frac{1}{2}\right) & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & 0 & 0 & 1 \end{pmatrix} && \left(R_2 \leftarrow R_2 - \left(\frac{1}{2}\right) R_1\right) \\
&\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & 1 & 0 & 0 \\ 0 & \frac{\frac{1}{4}}{3} & \frac{\frac{1}{3}}{4} & -\left(\frac{1}{2}\right) & 1 & 0 \\ 0 & \frac{\frac{1}{3}}{4} & \frac{\frac{4}{9}}{5} & -\left(\frac{1}{3}\right) & 0 & 1 \end{pmatrix} && \left(R_3 \leftarrow R_3 - \left(\frac{1}{3}\right) R_1\right) \\
&\sim \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & 1 & 0 & 0 \\ 0 & 1 & \frac{\frac{1}{3}}{4} & \frac{-\left(\frac{1}{2}\right)}{\frac{1}{4}} & \frac{\frac{1}{1}}{4} & 0 \\ 0 & \frac{\frac{1}{3}}{4} & \frac{\frac{4}{9}}{5} & -\left(\frac{1}{3}\right) & 0 & 1 \end{pmatrix} && \left(R_2 \leftarrow \frac{1}{\frac{1}{4}} R_2\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{-\frac{1}{2}}{3} & 4 & -6 & 0 \\ 0 & 1 & \frac{\frac{1}{3}}{4} & \frac{-\left(\frac{1}{2}\right)}{\frac{1}{4}} & \frac{\frac{1}{1}}{4} & 0 \\ 0 & \frac{\frac{1}{3}}{4} & \frac{\frac{4}{9}}{5} & -\left(\frac{1}{3}\right) & 0 & 1 \end{pmatrix} && \left(R_1 \leftarrow R_1 - \left(\frac{1}{2}\right) R_2\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{-\frac{1}{2}}{3} & 4 & -6 & 0 \\ 0 & 1 & \frac{\frac{1}{3}}{4} & \frac{-\left(\frac{1}{2}\right)}{\frac{1}{4}} & \frac{\frac{1}{1}}{4} & 0 \\ 0 & 0 & \frac{\frac{1}{36}}{5} & \frac{1}{6} & -1 & 1 \end{pmatrix} && \left(R_3 \leftarrow R_3 - \left(\frac{\frac{1}{3}}{4}\right) R_2\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{-\frac{1}{2}}{3} & 4 & -6 & 0 \\ 0 & 1 & \frac{\frac{1}{3}}{4} & \frac{-\left(\frac{1}{2}\right)}{\frac{1}{4}} & \frac{\frac{1}{1}}{4} & 0 \\ 0 & 0 & 1 & \frac{\frac{1}{6}}{\frac{1}{36}} & \frac{-1}{\frac{1}{36}} & \frac{1}{\frac{1}{36}} \end{pmatrix} && \left(R_3 \leftarrow \frac{1}{\frac{\frac{1}{36}}{5}} R_3\right) \\
&\sim \begin{pmatrix} 1 & 0 & 0 & 9 & -36 & 30 \\ 0 & 1 & \frac{\frac{1}{3}}{4} & \frac{-\left(\frac{1}{2}\right)}{\frac{1}{4}} & \frac{\frac{1}{1}}{4} & 0 \\ 0 & 0 & 1 & \frac{\frac{1}{6}}{\frac{1}{36}} & \frac{-1}{\frac{1}{36}} & \frac{1}{\frac{1}{36}} \end{pmatrix} && \left(R_1 \leftarrow R_1 - \left(\frac{-\frac{1}{2}}{3}\right) R_3\right) \\
&\sim \begin{pmatrix} 1 & 0 & 0 & 9 & -36 & 30 \\ 0 & 1 & 0 & \frac{-3}{\frac{1}{4}} & \frac{16}{\frac{1}{4}} & -180 \\ 0 & 0 & 1 & \frac{\frac{1}{6}}{\frac{1}{36}} & \frac{-1}{\frac{1}{36}} & \frac{1}{\frac{1}{36}} \end{pmatrix} && \left(R_2 \leftarrow R_2 - \left(\frac{\frac{\frac{1}{3}}{4}}{\frac{1}{4}}\right) R_3\right) \\
H(3)^{-1} &= \begin{pmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -180 & 180 \end{pmatrix}
\end{aligned}
$$

Symbolic entries and matrix-valued functions

An explicit dimension lets the function parameter remain inside the entries:

$$S(t) = \left(t^{i+j-2}\right)_{i,j=1}^{2}.$$

$$S\left(t\right) = \begin{pmatrix} 1 & t \\ t & t^2 \end{pmatrix}$$

The rows are symbolically dependent, so the determinant vanishes:

$$\begin{aligned} S\left(t\right) &= \begin{pmatrix} 1 & t \\ t & t^2 \end{pmatrix} \\ &\sim \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix} \quad (R_2 \leftarrow R_2 - (t)\, R_1) \\ \det\left(S\left(t\right)\right) &= 0 \end{aligned}$$

Matrix limits are taken entry by entry:

$$\lim_{t \to 0} S\left(t\right) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

A rectangular rule supplies both dimensions explicitly:

$$(10i + j)_{i,j}^{2\times 3} = \begin{pmatrix} 11 & 12 & 13 \\ 21 & 22 & 23 \end{pmatrix}$$

Limits through linear algebra

For the averaging matrix P, powers converge to projection onto the steady state.

$$\lim_{n \to \infty} P^n = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

Continuous linear-algebra operations consume the same symbolic sequence:

$$\lim_{n \to \infty} \operatorname{tr}\left(P^n\right) = 1$$
$$\lim_{n \to \infty} \det\left(P^n\right) = 0$$

The source definition, exact calculations, and displayed matrices above are all evaluated by SCAS while the document is built.